

- huber loss
1. Bellman equation
 2. ~~ReLU~~
 3. G-learning
 4. forget gate
 5. Regional Proposal semantic segmentation
 - 6.
 7. $B^2 + V$
 8. $(10 \times 10 \times 3 + 1) \times 5 = 1505$
 9. $\frac{128-10}{2} + 1 = 60$
 10. 2
 11. DBSCAN
 12. faster R-CNN
 13. 1
 14. policy gradient
 15. backpropagation
 16. non-max suppression
 17. Q-function
 18. boundary
 19. P x Q
 20. adaptive momentum

1. T
2. F
3. T
4. T
5. T
6. T
7. F
8. F
9. T
10. F

1. cell: $20 \times (-1) + 5 \times 3 + 30 \times (-2) + 0 \times 2 = -65$
- 1) the convolved image feature map:
- 2) applied ReLU
- 3) applied avg filter.

-65	730
90	-28
-52	140

0	730
90	0
0	140

365
45
70

2. forward pass

$$\text{net}_{h1} = 2 \times 0.4 + (-1) \times (-0.1) = 0.9, \quad \text{sigmoid}(\text{net}_{h1}) = \frac{1}{1+e^{-0.9}} = 0.7109, \quad \text{net}_{o1} = 0.7109 \times 0.5 + 0.8176 \times (-0.2) = 0.19195$$

$$\text{net}_{h2} = 2 \times 0.6 + (-1) \times (-0.3) = 1.5, \quad \text{sigmoid}(\text{net}_{h2}) = \frac{1}{1+e^{-1.5}} = 0.8176, \quad \text{sigmoid}(\text{net}_{o1}) = 0.5478$$

backpropagation, $y_1 = 1$, learning rate = 1.

$$\delta_o^1 = (o_1 - y_1) \cdot o_1(1-o_1) = (0.5478 - 1) \times 0.5478 \times (1-0.5478) = -0.111995$$

$$\delta_{h1}^1 = o_{h1}(1-o_{h1}) \cdot \delta_o^1 \cdot W_{oh1} = 0.7109 \times (1-0.7109) \times 0.5 \times (-0.111995) = -0.0115$$

$$\delta_{h2}^1 = o_{h2}(1-o_{h2}) \cdot \delta_o^1 \cdot W_{oh2} = 0.8176 \times (1-0.8176) \times (-0.2) \times (-0.111995) = 0.00334$$

$$W_{ho1} = 0.5 - (-0.111995) \times 0.7109 = 0.5796, \quad W_{ho2} = -0.2 - (-0.111995) \times 0.8176 = -0.1084$$

$$W_{xh1} = 0.4 - (-0.0115) \times 2 = 0.423, \quad W_{xh2} = 0.6 - (-0.0115) \times 2 = 0.5933$$

$$W_{xh1} = -0.1 - (-0.0115) \times (-1) = -0.9885, \quad W_{xh2} = -0.3 - 0.00334 \times (-1) = -0.29666$$

$$3. \quad W_{xf} = 0.1, \quad W_{hf} = 2.3, \quad b_f = 2.1 \quad x^t = -2.5, \quad h^{t-1} = -0.8, \quad t^t = 1.2$$

$$W_{xi} = -1.4, \quad W_{hi} = -1.1, \quad b_i = -1.7 \Rightarrow i^t = 0(W_{xi} \cdot x^t + W_{hi} \cdot h^{t-1} + b_i) = 0(2.68) = 0.9358$$

$$W_{xc} = 0.9, \quad W_{hc} = -0.5, \quad b_c = 1.8 \quad c^t = \tanh(W_{xc} \cdot x^t + W_{hc} \cdot h^{t-1} + b_c) = \tanh(-0.05) = -0.0499$$

$$W_{xo} = -0.8, \quad W_{ho} = 1.6, \quad b_o = -2.6 \quad o^t = f(i^t, c^t) = i^t \cdot c^t = 0.502 \times 1.2 + 0.9358 \times (-0.0499) = 0.556$$

$$h^t = 0^t \cdot \tanh(c^t) = 0.1323 \times \tanh(0.556) = 0.06688$$

$$4. \quad P(\text{the}+) = \frac{3}{8} = 0.375, \quad P(\text{the}^-) = \frac{3}{8} = 0.375, \quad P(\text{the}^0) = \frac{2}{8} = 0.25$$

positive this movie is amazing theater very clean with ideal

negative the movie price is very costly theater food not amazing crowded for this → 16 words

neutral the food at this theater is not costly afternoon ideal for movie → 14 words

$$P(\text{the}+) = \frac{0+1}{16+18} = 0.294, \quad P(\text{movie}+) = \frac{2+1}{16+18} = 0.882, \quad P(\text{theater}+) = \frac{2+1}{16+18} = 0.088, \quad P(\text{is}+) = \frac{3+1}{16+18} = 0.1176$$

$$P(\text{costly}+) = \frac{0+1}{16+18} = 0.294, \quad P(\text{the}^-) = \frac{2+1}{19+18} = 0.081, \quad P(\text{movie}^-) = \frac{2+1}{19+18} = 0.081, \quad P(\text{theater}^-) = \frac{2+1}{19+18} = 0.081$$

$$P(\text{is}^-) = \frac{3+1}{19+18} = 0.108, \quad P(\text{costly}^-) = \frac{1+1}{19+18} = 0.054$$

~~variance~~

$$P(\text{the}|\sim) = \frac{1+1}{14+18} = 0.0625, P(\text{movie}|\sim) = \frac{1+1}{14+18} = 0.0625, P(\text{theatre}|\sim) = \frac{1+1}{14+18} = 0.0625.$$

$$P(\text{is}|\sim) = \frac{2+1}{14+18} = 0.09375, P(\text{costly}|\sim) = \frac{1+1}{14+18} = 0.0625$$

$$P(\text{instance}|\sim) = 0.375 \times 0.294 \times 0.882 \times 0.088 \times 0.1176 \times 0.294 = 0.00295859$$

$$P(\text{instance}|\sim) = 0.375 \times 0.081 \times 0.081 \times 0.081 \times 0.108 \times 0.054 = 0.00000194$$

$$P(\text{instance}|\sim) = 0.25 \times 0.0625 \times 0.09375 = 0.00000036$$

$$P(\text{sentiment}|\sim) \propto P(\text{sentiment}, \text{input}), \text{argmax} \{ +NB, -NB, \sim NB \} = +NB.$$

$$P(+|\sim) = \frac{0.00295859}{0.00295859 + 0.00000194 + 0.00000036} = 0.9923$$

$$50 P(\text{Yes}) = \frac{8}{14} = 0.5714, P(\text{No}) = \frac{6}{14} = 0.4285$$

$$P(\text{dog}|\text{yes}) = \frac{1+1}{8+3} = 0.4545, P(\text{small}|\text{yes}) = \frac{3+1}{8+3} = 0.3636, P(\text{white}|\text{yes}) = \frac{3+1}{8+3} = 0.3636$$

$$P(\text{dog}|\text{no}) = \frac{1+1}{6+3} = 0.22, P(\text{small}|\text{no}) = \frac{2+1}{6+3} = 0.333, P(\text{white}|\text{no}) = \frac{2+1}{6+3} = 0.333$$

$$P(\text{yes}|\text{input}) \propto P(\text{yes}, \text{input}) = 0.5714 \times 0.4545 \times 0.3636 \times 0.3636 = 0.0343$$

$$P(\text{no}|\text{input}) \propto P(\text{no}, \text{input}) = 0.4285 \times 0.22 \times 0.333 \times 0.333 = 0.01047$$

$$\text{argmax} \{ \text{yesNB}, \text{noNB} \} = \text{yesNB}, P(\text{yes}|\text{input}) = \frac{0.0343}{0.0343 + 0.01047} = 0.7654$$

$$2 P(\text{cow}|\text{yes}) = \frac{2+1}{8+3} = 0.2727, P(\text{medium}|\text{yes}) = \frac{2+1}{8+3} = 0.2727, P(\text{brown}|\text{yes}) = \frac{2+1}{8+3} = 0.2727$$

$$P(\text{cow}|\text{no}) = \frac{2+1}{6+3} = 0.333, P(\text{medium}|\text{no}) = \frac{2+1}{6+3} = 0.333, P(\text{brown}|\text{no}) = \frac{3+1}{6+3} = 0.444$$

$$P(\text{yes}|\text{input}) \propto P(\text{yes}, \text{input}) = 0.5714 \times 0.2727 \times 0.2727 = 0.01159$$

$$P(\text{no}|\text{input}) \propto P(\text{no}, \text{input}) = 0.4285 \times 0.333 \times 0.444 = 0.021097$$

$$\text{argmax} \{ \text{yesNB}, \text{noNB} \} = \text{noNB}, P(\text{no}|\text{input}) = \frac{0.021097}{0.021097 + 0.01159} = 0.6454$$